

NQS-Updates

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Energy witnesses only work for ground states, since they need H to have the target as its minimum. The Gaussian witness of [arXiv:1703.03152](#) instead relies on the state being Gaussian, which is a property a time-evolved state keeps. That is why it can certify a quench, and why an energy witness cannot.

1 Direct Certification of Quantum Simulations

Reference: Hangleiter, Kliesch, Schwarz and Eisert, [arXiv:1602.00703](#).

The problem this paper addresses: a quantum simulator is worth building only when it can do something classically intractable, but then one cannot check it by computing the answer classically. The resolution is that certification does not require *predicting* the device’s output, only *verifying a property* of the state it produced — and the energy is such a property, since it is locally measurable and tied to the fidelity through the spectral gap.

To clarify.

- BQP, and how it relates to NP and to the IQP circuits used later.
- The precise content of the fidelity / trace-distance relation (Fuchs–van de Graaf inequalities): why both bounds are needed, and which direction the certification argument uses.

1.1 Weak-membership certification

Since we cannot perfectly certify that F is larger than or equal to a threshold F_T — a finite number of measurements can never decide the borderline case — the guarantee is stated with a buffer. Arthur’s test must:

- **accept** if $F \geq F_T + \delta$, with probability $\geq 1 - \alpha$;
- **reject** if $F \leq F_T$, with probability $\geq 1 - \alpha$;
- give **no promise** in between.

Here δ is the *fidelity gap*: the width of the inconclusive band. The output of the protocol is therefore not an estimate “the fidelity is X ”, but the statement “I accept, and acceptance implies $F \geq F_T$ ”.

1.2 Structural assumptions

Three conditions carry the result, and each is used at a different point.

- (i) **k -locality:** $H = \sum_{i=1}^n h_i$ with each h_i acting on $O(1)$ sites and $\|h_i\| \leq J$. Needed so that Arthur can measure each h_i with a local measurement.
- (ii) **Frustration-freeness:** every local term annihilates the ground state, $h_i|\psi_0\rangle = 0$, which pins $E_0 = 0$. This is the load-bearing assumption: it means Arthur needs *no prior knowledge of the ground energy*, since the measured $\langle H \rangle$ is by construction the energy *above* the ground state. In a generic Hamiltonian one would first have to compute E_0 , which is the hard problem being avoided.
- (iii) **Polynomial spectral gap:** $\Delta = \Delta(N) \geq 1/\text{poly}(N)$. The gap converts energy error into fidelity error and appears as $1/\Delta$ in the bound and $1/\Delta^2$ in the sample count, so an exponentially small gap would destroy the scheme while a polynomially small one only degrades it gradually.

1.3 Lemma 2

The mathematical core of the paper is the two-sided bound

$$1 - \frac{\langle H \rangle_\varrho}{\Delta} \leq F(\varrho_0, \varrho) \leq 1 - \frac{\langle H \rangle_\varrho}{\|H\|}. \quad (1)$$

Derivation. Diagonalise $H = \sum_k E_k |k\rangle\langle k|$ with $E_0 = 0$ (frustration-freeness) and $E_1 = \Delta$. For a state ϱ write its populations in the energy basis as $p_k = \langle k|\varrho|k\rangle$; these are non-negative because $\varrho \succeq 0$, and sum to 1 because $\text{tr } \varrho = 1$. Since the target $\varrho_0 = |0\rangle\langle 0|$ is pure,

$$F = \text{tr}[\varrho_0 \varrho] = p_0, \quad \text{hence} \quad 1 - F = \sum_{k \geq 1} p_k, \quad (2)$$

so the infidelity is exactly the total population on excited states. The energy involves the same populations, with the $k = 0$ term absent because $E_0 = 0$:

$$\langle H \rangle_\varrho = \sum_{k \geq 1} E_k p_k. \quad (3)$$

Both quantities are therefore sums over the same set $\{p_k\}_{k \geq 1}$, differing only in the weight: 1 for the infidelity, E_k for the energy. Every excited eigenvalue satisfies $\Delta \leq E_k \leq \|H\|$, so multiplying by $p_k \geq 0$ and summing over $k \geq 1$ gives

$$\Delta(1 - F) \leq \langle H \rangle_\varrho \leq \|H\|(1 - F), \quad (4)$$

and dividing by $\Delta > 0$ and $\|H\| > 0$ respectively yields the two bounds.

Role of each side. The lower bound $F_{\min} = 1 - \langle H \rangle / \Delta$ is the *certificate*: a low measured energy forces a high fidelity, because each unit of misplaced probability costs at least Δ in energy. It is always valid but informative only when $\langle H \rangle < \Delta$; below that it turns negative, exactly as the witness in the previous section does at small β . The upper bound provides *soundness*: a high energy forces a low fidelity, so a bad state cannot fake a good energy reading, which is what allows Arthur to reject. Both directions are needed — without the upper bound the test could accept everything.